User Guide
Plugin Semantic Segmentation

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The plugin works with QGIS version 3.x. No additional software is required beyond QGIS and the plugin archive. The plugin requires approximately 3GB to install its dependencies.

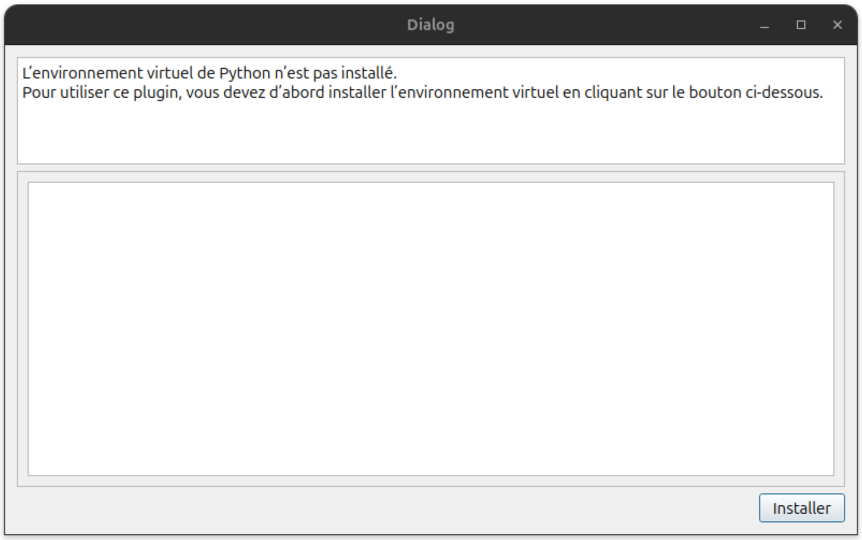
Installation

The first step is to download the plugin archive (if you do not have it, you can contact me at *kelian.pons@centrale.centralelille.fr*). Once downloaded, extract the archive into the QGIS plugin directory. To locate this directory, open QGIS and go to:

`Preferences > User Profiles > Open Active Profile Folder`

In this folder, navigate to `python/plugins/`. Extract the archive into this directory. Restart QGIS to make the plugin appear.

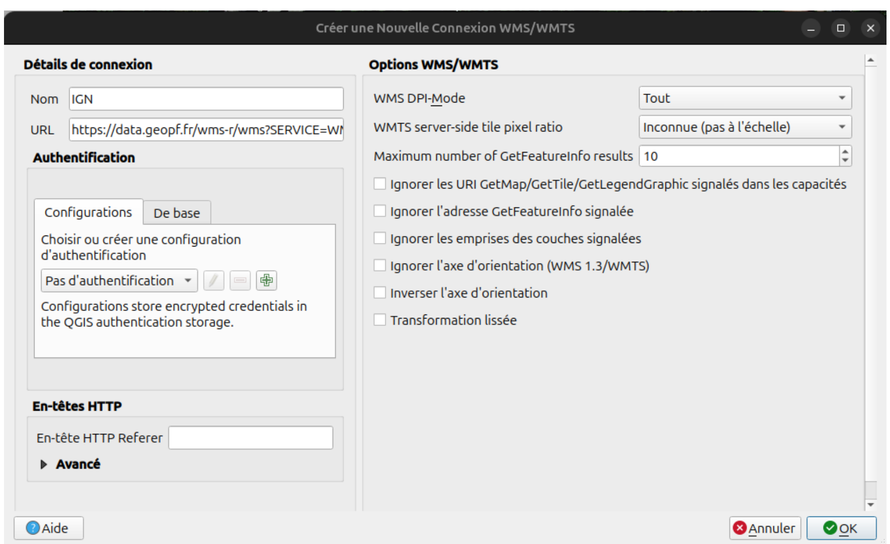
You can launch the plugin by clicking its icon in the toolbar (usually located near the Python console), or via the *Raster* menu. The first time you start the plugin, an installation window will appear. This step installs all required dependencies and may take several minutes.



Access to IGN Data

Before using the plugin, you may need to download data from IGN. To do so, follow these instructions (also available on the IGN website: https://cartes.gouv.fr/aide/fr/guides-utilisateur/utiliser-les-services-de-la-geoplateforme/tutoriels-api/qgis/).

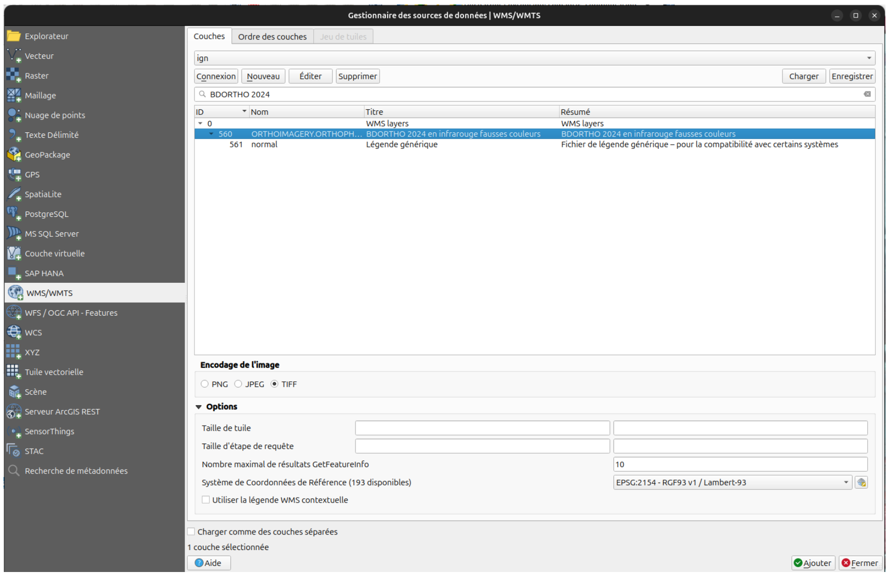
In QGIS, add a WMS connection. In the left panel (*Browser*), right-click on *WMS/WMTS* and select *New Connection*. The following window should appear:



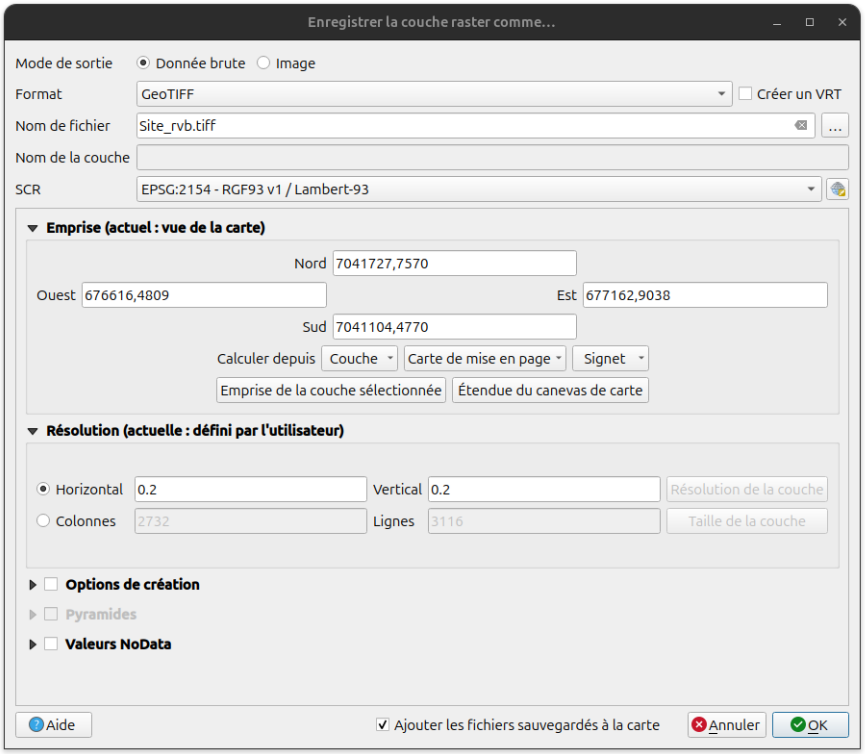
Enter any name you like and use the following URL: https://data.geopf.fr/wms-r/wms?

You can now add layers from the IGN database via this connection. The plugin requires at least the red, green, and blue bands (and optionally near-infrared for improved accuracy).

To add a layer, you can use the shortcut `Ctrl + Shift + W` or navigate to `Layer > Add a layer > Add a WMS/WMTS Layer`. Select the WMS connection you just added and click on Connection. Search for *photographies aériennes* and select imagery from 2021 or 2024 for the Pas-de-Calais department (for more information on other departments see Table 1). To obtain the near-infrared band, search for *infrarouge fausses couleurs* and select the same year.



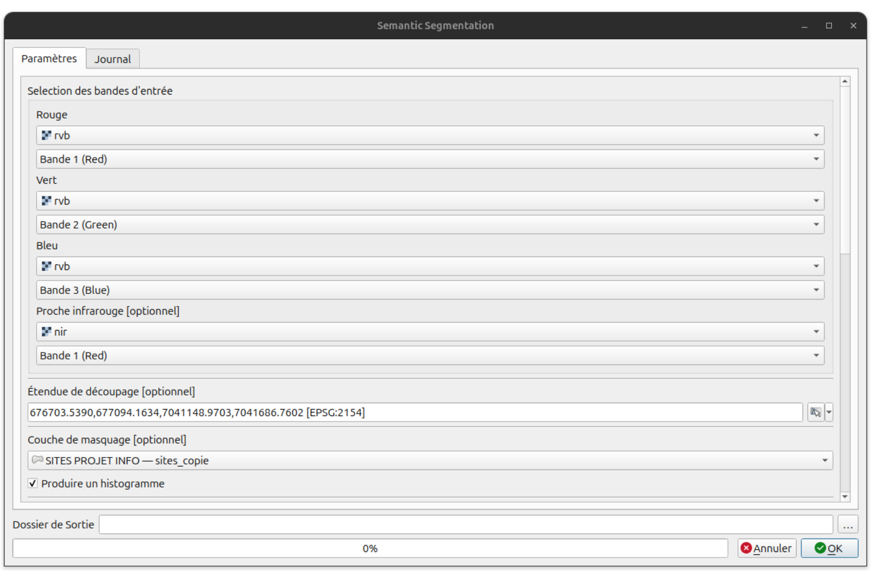
Once the data is loaded, you will need to export your area of interest. Use the current map view to define the area, then in the Layer Panel right-click the layer and select `Export > Save As...`. The following window will appear:



Uncheck the *Create a VRT* option and enter a file path and name with the extension *.tiff* or *.tif*. Be sure to select *Map extent*. Otherwise, the entire dataset (up to 50 GB) will be downloaded. Set both the horizontal and vertical resolution to 0.2 for orthophotos (same as the image).

1 Segmentation

You can now run the plugin. Launch it from the toolbar or the *Raster* menu.



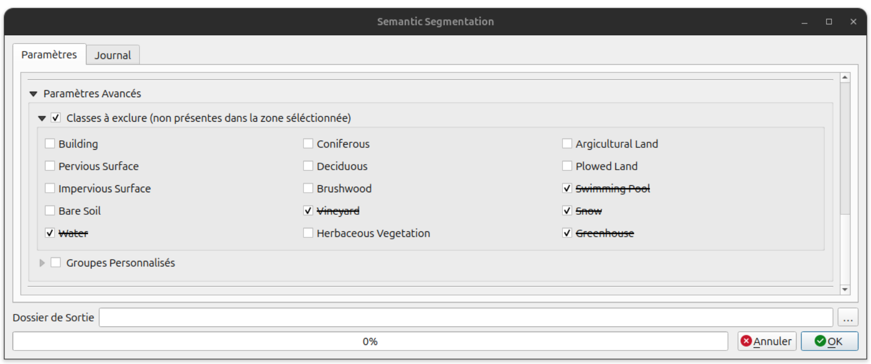
Select the input bands (red, green, blue, and near-infrared). You may also define:

- An extent (applied before the segmentation).
- A mask layer (applied after the segmentation).

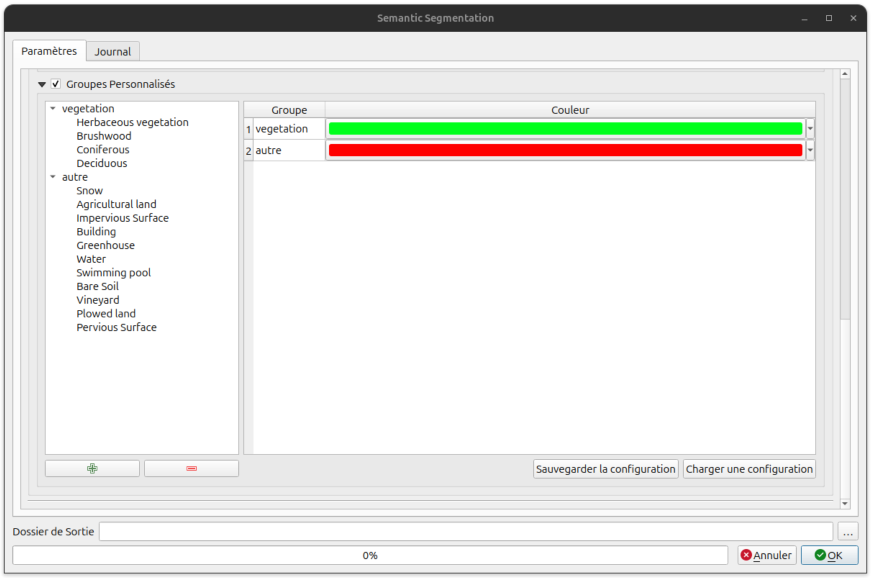
The option *Produce histogram* is only taken into account if a mask layer is provided.

You can choose between two output modes:

Class Exclusion If you know that a class is absent from the image, you can exclude it. This does not modify the model itself, but only the post-processing. The model outputs, for each pixel, probabilities for each class. The excluded class probabilities are set to zero, and the final label corresponds to the highest remaining probability.

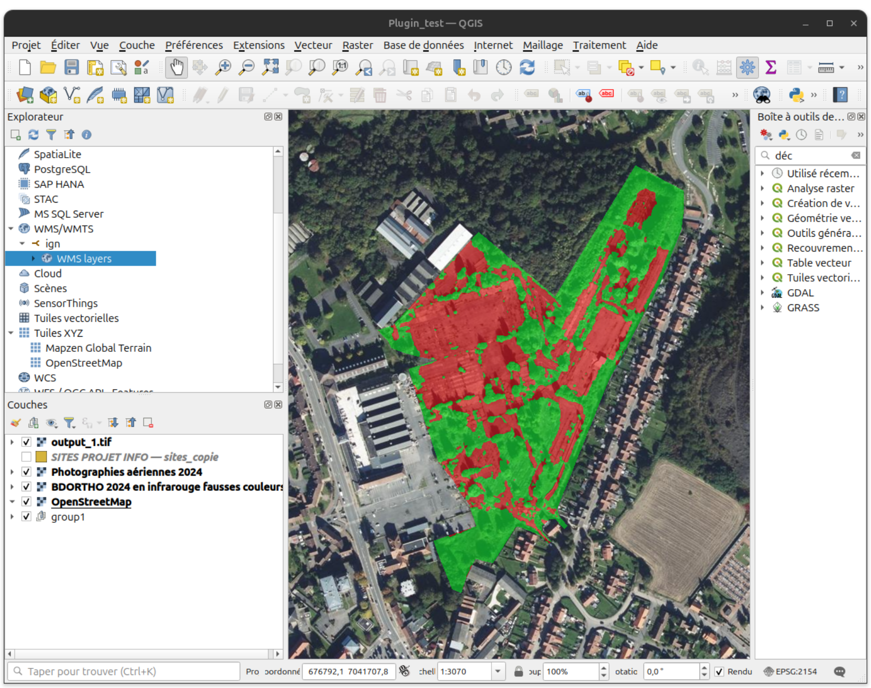


Class Grouping This option provides more flexibility by allowing you to group classes (e.g., vegetation), combining categories such as coniferous, deciduous, and herbaceous vegetation. To create a new group, you can simply click on the button and enter a name for your new group. You can also delete a group with the button. Configurations can be saved, and you need to save them under JSON format (.json). You can also load a configuration using the corresponding button. Avoid using special or accented characters for group names.



After configuring the parameters, select an output directory. You can then start the inference. Depending on the input size, processing typically takes 1 to 5 minutes. Progress can be monitored via the progress bar and logs.

If successful, the output layer will be automatically added to the current QGIS project.



2 Uninstallation

To uninstall the plugin, go to the plugin directory (`Preferences > User Profiles > Open Active Profile Folder`, then `python/plugins/`) and delete the `semantic_segmentation` folder.

Table 1: Department and year photographed

Number	Year	Department	Number	Year	Department
01	2024	Ain	53	2022	Mayenne
02	2024	Aisne	54	2022	Meurthe-et-Moselle
03	2025	Allier	55	2022	Meuse
04	2024	Alpes-de-Haute-Provence	56	2024	Morbihan
05	2025	Hautes-Alpes	57	2022	Moselle
06	2023	Alpes-Maritimes	58	2023	Nièvre
07	2023	Ardèche	59	2025	Nord
08	2022	Ardennes	60	2024	Oise
09	2025	Ariège	61	2023	Orne
10	2025	Aube	62	2024	Pas-de-Calais
11	2024	Aude	63	2025	Puy-de-Dôme
12	2022	Aveyron	64	2024	Pyrénées-Atlantiques
13	2023	Bouches-du-Rhône	65	2025	Hautes-Pyrénées
14	2023	Calvados	66	2024	Pyrénées-Orientales
15	2022	Cantal	67	2024	Bas-Rhin
16	2023	Charente	68	2024	Haut-Rhin
17	2024	Charente-Maritime	69	2023	Rhône
18	2023	Cher	70	2023	Haute-Saône
19	2023	Corrèze	71	2023	Saône-et-Loire
21	2023	Côte-d'Or	72	2022	Sarthe
22	2025	Côtes-d'Armor	73	2024	Savoie
23	2024	Creuse	74	2023	Haute-Savoie
24	2024	Dordogne	75	2024	Paris
25	2023	Doubs	76	2022	Seine-Maritime
26	2023	Drôme	77	2024	Seine-et-Marne
27	2022	Eure	78	2024	Yvelines
28	2023	Eure-et-Loir	79	2023	Deux-Sèvres
29	2024	Finistère	80	2024	Somme
2A	2024	Corse-du-Sud	81	2022	Tarn
2B	2024	Haute-Corse	82	2025	Tarn-et-Garonne
30	2024	Gard	83	2023	Var
31	2022	Haute-Garonne	84	2024	Vaucluse
32	2025	Gers	85	2022	Vendée
33	2024	Gironde	86	2023	Vienne
34	2024	Hérault	87	2024	Haute-Vienne
35	2023	Ille-et-Vilaine	88	2023	Vosges
36	2023	Indre	89	2023	Yonne
37	2025	Indre-et-Loire	90	2023	Territoire-de-Belfort
38	2024	Isère	91	2024	Essonne
39	2023	Jura	92	2024	Hauts-de-Seine
40	2024	Landes	93	2024	Seine-Saint-Denis
41	2023	Loir-et-Cher	94	2024	Val-de-Marne
42	2025	Loire	95	2024	Val-d'Oise
43	2025	Haute-Loire	971	2024	Guadeloupe
44	2025	Loire-Atlantique	972	2025	Martinique
45	2023	Loiret	973	2021	Guyane
46	2025	Lot	974	2022	Réunion
47	2024	Lot-et-Garonne	975	2022	Saint-Pierre-et-Miquelon
48	2025	Lozère	976	2023	Mayotte
49	2025	Maine-et-Loire	977	2024	Saint-Barthélemy
50	2025	Manche	978	2024	Saint-Martin
51	2022	Marne			
52	2022	Haute-Marne			